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1 % =====
2 % HVDC - four-terminal scheme
3 %
4 % Developed by: F. Bianchi, E. Prieto, S. Matta
5 %% =====
6 close all, clear all
7 fpath = './DCtech';
8
9 % =====
10 %% Network values
11 C1 = 150e-6;
12 C2 = 150e-6;
13 C3 = 150e-6;
14 C4 = 150e-6;
15 R1 = 0.5;
16 R2 = 0.25;
17 R3 = 0.4;
18 L1 = 0.005;
19 L2 = 0.0025;
20 L3 = 0.004;
21
22 % State-space model
23 A = [0, 0, 0, 0, -1/C1, -1/C1, 0;
24      0, 0, 0, 0, 0, 1/C2, -1/C2;
25      0, 0, 0, 0, 1/C3, 0, 0;
26      0, 0, 0, 0, 0, 0, 1/C4;
27      1/L1, 0, -1/L1, 0, -R1/L1, 0, 0;
28      1/L2, -1/L2, 0, 0, 0, -R2/L2, 0;
29      0, 1/L3, 0, -1/L3, 0, 0, -R3/L3];
30 B = [1/C1, 0, 0, 0;
31      0, 1/C2, 0, 0;
32      0, 0, -1/C3, 0;
33      0, 0, 0, -1/C4;
34      0, 0, 0, 0;
35      0, 0, 0, 0;
36      0, 0, 0, 0];
37 C = [1, 0, 0, 0, 0, 0, 0;
38      0, 1, 0, 0, 0, 0, 0;
39      0, 0, 1, 0, 0, 0, 0;
40      0, 0, 0, 1, 0, 0, 0];
41 D = zeros(4);
42
43 Bw = B(:,1:2); Bu = B(:,3:4);
44 Cz = C(1:2,:); Cy = C(3:4,:);
45
46 % =====
47 % Open loop transfers
48 G = ss(A,B,C,D);
49
50 Gzw = G(1:2,1:2); zpk(Gzw);
51 Gzu = G(1:2,3:4); zpk(Gzu);
52 Gyw = G(3:4,1:2); zpk(Gyw);
53 Gyu = G(3:4,3:4); zpk(Gyu);
54
55 % gains
56 Kgi = [100 50 20 7 3 1];
57 Kg = 1./Kgi;
58 for ii=1:length(Kgi)

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59     lgtext{ii} = sprintf('1/%d',Kg(ii));
60 end
61 lgtext{6} = '1';
62 P = diag([1 1]);
63 % colors
64 ColorSet = diag(linspace(0,0.5,2))*ones(2,3);
65 LineSet = {'-','--','-.'};
66
67 % =====
68 %% Stability
69 %
70 Hf = figure('Name','Closed loop poles');
71 Ha = axes('ColorOrder', ColorSet,...
72     'LineWidth',1,...
73     'Fontname','times',...
74     'FontSize',14,...
75     'NextPlot','add');
76
77 %for ii=1:2:length(Kg)
78 ii = 2;
79 eiAcl(:,ii) = eig(A+Kg(ii)*Bu*P*Cy);
80 plot(real(eiAcl(:,ii)),imag(eiAcl(:,ii)),'x',...
81     'MarkerEdgeColor',ColorSet(1,:),...
82     'MarkerFaceColor',ColorSet(1,:),...
83     'LineWidth',2,...
84     'MarkerSize',10)
85 ii = 4;
86 eiAcl(:,ii) = eig(A+Kg(ii)*Bu*P*Cy);
87 plot(real(eiAcl(:,ii)),imag(eiAcl(:,ii)),'o',...
88     'MarkerEdgeColor',ColorSet(1,:),...
89     'MarkerFaceColor',0.7*[1 1 1],...
90     'LineWidth',2,...
91     'MarkerSize',10)
92 ii = 6;
93 eiAcl(:,ii) = eig(A+Kg(ii)*Bu*P*Cy);
94 plot(real(eiAcl(:,ii)),imag(eiAcl(:,ii)),'s',...
95     'MarkerEdgeColor',ColorSet(1,:),...
96     'MarkerFaceColor',0.7*[1 1 1],...
97     'LineWidth',2,...
98     'MarkerSize',10)
99 plot([0 0],3000*[-1 1],'k')
100 plot([-7000 0],[0 0],'k')
101 % end
102 % ylabel('Imag(\lambda(A_{cl}))'); xlabel('Real(\lambda(A_{cl}))');
103 ylabel('Imag(l(Acl))'); xlabel('Real(l(Acl))');
104 legend(lgtext{[2 4 6]}, 'Location','SouthWest'), legend boxoff
105 print('-depsc2',[fpath 'fig_eigAcl_c1.eps'])
106
107 % det(1+L(jw))
108 % w = logspace(-1,4,2000);
109 % LL1 = frd(eye(2)+Gyu,w); rLL1 = frdata(LL1);
110 % LL2 = frd(eye(2)+5*Gyu,w); rLL2 = frdata(LL2);
111 % for ii=1:length(w)
112 %     dl1(ii) = det(rLL1(:, :, ii));
113 %     dl2(ii) = det(rLL2(:, :, ii));
114 % end
115 % plot(real(dl1),imag(dl1))
116 % hold

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117 % plot(real(d12),imag(d12),'r')
118
119
120 % =====
121 %% Closed loop transfers
122 systemnames = 'G';
123 inputvar = '[w{2};r{2};u{2}]';
124 outputvar = '[G(1:2);-r+G(3:4);-r+G(3:4)]';
125 input_to_G = '[w;u]';
126 G2 = sysic;
127
128 clear Gcl
129 for ii=1:length(Kg)
130     Gcl(:, :, ii) = lft(G2, Kg(ii)*P);
131 end
132
133 % frequencies
134 w = 2*pi*logspace(-1,3,1000);
135
136 % -----
137 %% Figure: sensitivity
138 S = Gcl(3:4,3:4,:);
139 %
140 Hf = figure('Name','S');
141 Ha = axes('ColorOrder', ColorSet, 'LineStyleOrder', LineSet, ...
142     'LineWidth',1,...
143     'Fontname','times',...
144     'FontSize',14,...
145     'NextPlot','add',...
146     'XScale','log',...
147     'Xlim',[1e-1 1e3], 'Ylim',[-40 20]);
148 for ii=1:length(Kg)
149     sv = sigma(S(:, :, ii), w);
150     svSmx(:, ii) = 20*log10(sv(1, :));
151     svSmn(:, ii) = 20*log10(sv(2, :));
152 end
153 semilogx(w/2/pi, svSmx)
154 semilogx(w/2/pi, svSmn, '--')
155 ylabel('SVD (dB)'); xlabel('Frequency (Hz)');
156 legend(lgtext), legend boxoff
157 % print('-depsc2', 'fig_svS_c1.eps')
158
159 % -----
160 %% Figure: SGyw
161 SGyw = Gcl(3:4,1:2,:);
162 Hf = figure('Name','SGyw');
163 Ha = axes('ColorOrder', ColorSet, 'LineStyleOrder', LineSet, ...
164     'Fontname','times',...
165     'FontSize',14,...
166     'XScale','log',...
167     'NextPlot','add',...
168     'Xlim',[1e-1 1e3], 'Ylim',[-20 50]);
169 for ii=1:length(Kg)
170     sv = sigma(SGyw(:, :, ii), w);
171     svSGyw(:, ii) = 20*log10(sv(1, :));
172 end
173 % limits
174 emax = 20*log10(22.5);

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175 h = area([0.1 200 2000],[emax emax emax+20]);
176 set(h,'FaceColor',[1 1 1]*0.9,'BaseValue',-20,'LineWidth',0.5)
177 %
178 H1 = semilogx(w/2/pi,svSGyw,'LineWidth',1);
179 set(H1(3),'LineWidth',2);
180 ylabel('SVD (dB)'); xlabel('Frequency (Hz)');
181 legend(H1,lgtext,'Location','SouthWest'), legend boxoff
182 print('-depsc2',[fpath 'fig_svSGyw_c1.eps'])
183 % svd(freqresp(Gyu,0)\freqresp(Gyw,0))
184 % dcgain
185 [Nyu,d] = tfdata(Gyu); Nyu = [Nyu{1,1}(end) Nyu{1,2}(end); Nyu{2,1}(end) Nyu{2,2}(end)];
186 [Nyw,d] = tfdata(Gzu); Nyw = [Nyw{1,1}(end) Nyw{1,2}(end); Nyw{2,1}(end) Nyw{2,2}(end)];
187
188 % -----
189 %% Figure: KSGyw
190 for ii=1:length(Kg)
191     KSGyw(:, :, ii) = Kg(ii)*SGyw(:, :, ii);
192 end
193 Hf = figure('Name','KSGyw');
194 Ha = axes('ColorOrder', ColorSet, 'LineStyleOrder', LineSet, ...
195     'Fontname','times', ...
196     'FontSize',14, ...
197     'NextPlot','add', ...
198     'XScale','log', ...
199     'Xlim',[1e-1 1e3], 'Ylim',[-40 20]);
200 for ii=1:length(Kg)
201     sv = sigma(KSGyw(:, :, ii),w);
202     svKSGyw(:, ii) = 20*log10(sv(1,:));
203 end
204 % limits
205 emax = 20*log10(1.50);
206 h = area([0.1 200 2000],[emax emax emax-20]);
207 set(h,'FaceColor',[1 1 1]*0.9,'BaseValue',-40,'LineWidth',0.5)
208 %
209 H1 = semilogx(w/2/pi,svKSGyw,'LineWidth',1);
210 set(H1(3),'LineWidth',2);
211 ylabel('SVD (dB)'); xlabel('Frequency (Hz)');
212 legend(H1,lgtext,'Location','SouthWest'), legend boxoff
213 print('-depsc2',[fpath 'fig_svKSGyw_c1.eps'])
214
215 % -----
216 %% Figure: GzuKS
217 GzuKS = Gcl(1:2,3:4,:);
218 Hf = figure('Name','GzuKS');
219 Ha = axes('ColorOrder', ColorSet, 'LineStyleOrder', LineSet, ...
220     'LineWidth',1, ...
221     'Fontname','times', ...
222     'FontSize',14, ...
223     'NextPlot','add', ...
224     'XScale','log', ...
225     'Xlim',[1e-1 1e3], 'Ylim',[-40 20]);
226 for ii=1:length(Kg)
227     sv = sigma(GzuKS(:, :, ii),w);
228     svGzuKS(:, ii) = 20*log10(sv(1,:));
229 end
230 H1 = semilogx(w/2/pi,svGzuKS,'LineWidth',1);
231 ylabel('SVD (dB)'); xlabel('Frequency (Hz)');
232 set(H1(3),'LineWidth',2);

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233 legend(H1,lgttext,'Location','SouthWest'), legend boxoff
234 % print('-depsc2','fig_svGzuKS_c1.eps')
235 % dcgain
236 [Nyu,d] = tfdata(Gyu); Nyu = [Nyu{1,1}(end) Nyu{1,2}(end); Nyu{2,1}(end) Nyu{2,2}(end)];
237 [Nzu,d] = tfdata(Gzu); Nzu = [Nzu{1,1}(end) Nzu{1,2}(end); Nzu{2,1}(end) Nzu{2,2}(end)];
238 % svd(freqresp(Gzu,0)/freqresp(Gyu,0))
239
240 % -----
241 %% Figure: Gzw_GzuKSGyw
242 Gzw_GzuKSGyw = Gcl(1:2,1:2,:);
243 Hf = figure('Name','Gzw_GzuKSGyw');
244 Ha = axes('ColorOrder', ColorSet,'LineStyleOrder', LineSet,...
245     'LineWidth',1,...
246     'Fontname','times',...
247     'FontSize',14,...
248     'NextPlot','add',...
249     'XScale','log',...
250     'Xlim',[1e-1 1e3],'Ylim',[-20 50]);
251 for ii=1:length(Kg)
252     sv = sigma(Gzw_GzuKSGyw(:, :, ii), w);
253     svGzw_GzuKSGyw(:, ii) = 20*log10(sv(1, :));
254 end
255 % limits
256 emax = 20*log10(22.5);
257 h = area([0.1 200 2000],[emax emax emax+20]);
258 set(h,'FaceColor',[1 1 1]*0.9,'BaseValue',-20,'LineWidth',0.5)
259 %
260 H1 = semilogx(w/2/pi,svGzw_GzuKSGyw,'LineWidth',1);
261 set(H1(3),'LineWidth',2);
262 ylabel('SVD (dB)'); xlabel('Frequency (Hz)');
263 legend(H1,lgttext,'Location','SouthWest'), legend boxoff
264 print('-depsc2',[fpath 'fig_svGzw_GzuKSGyw_c1.eps'])
265
266 %%
267 Kg = 1/22.5;
268 Gcl_0 = lft(G2,Kg*eye(2));
269 Gzw_GzuKSGyw = Gcl_0(1:2,1:2);
270 dcgain(Gzw_GzuKSGyw)*[667;667]
271 totalbus=dcgain(Gzw_GzuKSGyw)*[667;667]+[145e3;145e3]
272 %%
273 Kg = 1/20;
274 Gcl_0 = lft(G2,Kg*eye(2));
275 Gzw_GzuKSGyw = Gcl_0(1:2,1:2);
276 dcgain(Gzw_GzuKSGyw)*[633.2;633.2]
277 totalbus=dcgain(Gzw_GzuKSGyw)*[633.2;633.2]+[145e3;145e3]
278
279
280
281

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